Do Now · retrieval

Lesson 114 · Compilers, Interpreters and Assemblers

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

Does free-of-charge imply that software is open source?

Q2

A full adder receives A=1, B=1 and carry-in=0. What are sum and carry-out?

Q3

For the binary value 1010, what is its decimal value?

Q4

A linked-list node has data and what other essential field?

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Compiler — Translates a complete source unit to target code before execution, often enabling optimisation.

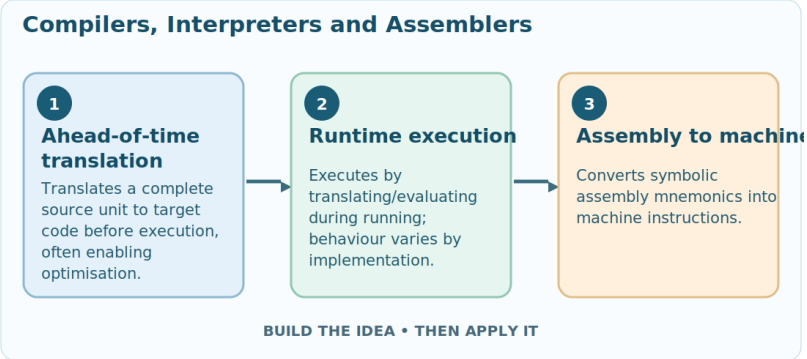
Core idea 2

Interpreter — Executes by translating/evaluating during running; behaviour varies by implementation.

Core idea 3

Assembler — Converts symbolic assembly mnemonics into machine instructions.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

compiler — Translates source to target code before execution.

interpreter — Executes by translating/evaluating at runtime.

assembler — Turns assembly into machine/object code.

object code — Compiled code awaiting linking or execution.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

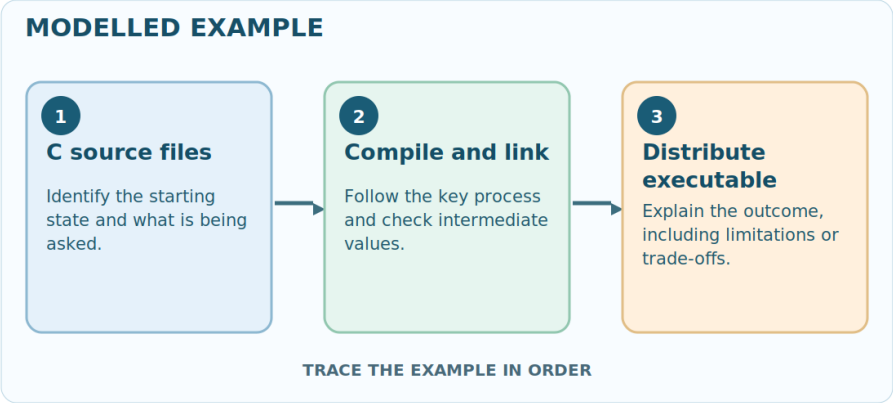
Problem

A team distributes an executable without its original C source. Explain the compiler's role.

Follow the worked method

1. Model answer — A compiler translates source to target machine/object code; linked executable can run without the source being present. Compilation can report syntax errors and apply optimisation before execution. — Check: link each claim to the scenario and use precise OCR terminology.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

What must we decide?
Identify the inputs, relevant facts and exact output or judgement required.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Distinguish compiler, interpreter and assembler with a relevant input for each. [6]

Question 2

Explain two advantages of compiling a program before distribution. [6]

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

A Python-like runtime, C compiler and LMC assembler are used in one project. Compare debugging, portability and execution trade-offs. [8]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

An assembler translates high-level Python syntax into binary machine code.

Use key vocabulary from memory

1. Explain compiler in your own words.

2. Explain interpreter in your own words.

3. Explain assembler in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.